

Revisiting RGB-D Fusion for Indoor Semantic Segmentation in the Era of Vision Foundation Models

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Abstract—Indoor RGB-D semantic segmentation has been revitalised by Vision Foundation Models (VFMs), yet it remains unclear how best to inject depth cues into these large pretrained backbones. Addressing this gap, we benchmark a DINOv2 ViT-L/14 baseline on NYUv2 and systematically compare early (input concatenation), mid-level (dual-stream feature concatenation) and late-level fusion schemes. A frozen backbone coupled to a single linear decoder already matches the depth-augmented DEADFL-UNet, substantiating the strong semantic prior of VFMs. Full fine-tuning lifts the RGB-only baseline to 69.9% mIoU, but naïve early or mid fusion of depth offers no further gain. In contrast, a lightweight late-fusion adapter that performs cross-modal attention (CMA) conditions RGB tokens on depth embeddings, raising performance to 71.4% mIoU. Targeted refinements—a staged backbone unfreezing schedule, a deeper convolutional depth encoder and a transposed-convolution decoder—push the score to 76.3% mIoU, surpassing the CNN benchmark by more than 17 percentage points while adding < 10% parameter overhead. These results reveal that depth remains valuable once it modulates, rather than merely merges with, high-level RGB features, and they position CMA-based fusion as a compact, state-of-the-art solution for indoor scene parsing. The study opens avenues for depth-to-RGB distillation and self-supervised depth utilisation in multimodal VFMs.

I. INTRODUCTION

Semantic segmentation is a fundamental task in computer vision, involving pixel-level classification to label each pixel in an image with its corresponding semantic category (1; 2). Unlike image classification, which predicts a single label per image, semantic segmentation provides a more granular understanding of visual content, segmenting an image into regions corresponding to objects, backgrounds, and other meaningful categories. Due to its detailed output, semantic segmentation is crucial for numerous practical applications, including autonomous driving, medical image analysis, robotic navigation, and augmented reality, where accurate scene understanding is essential for decision-making and interaction (3; 4).

Indoor semantic segmentation has traditionally relied on convolutional neural network architectures, notably U-Net-style models, which efficiently leverage RGB and depth data (RGB-D) for enhanced scene understanding (5; 6; 7). Tran et al. (8) exemplify this approach, employing a depth-augmented U-Net architecture optimized through fine-tuning to achieve state-of-the-art performance on the NYUv2 dataset (9). However, traditional convolutional architectures typically require extensive customization and optimization to effectively

fuse heterogeneous RGB and depth features, which can be resource-intensive and limit scalability (10; 4). Consequently, there is strong motivation to explore alternative methods, such as vision foundation models, to potentially reduce the need for extensive task-specific tuning while leveraging powerful generalized representations (11; 12).

Recent advances in vision foundation models (VFMs), especially transformer-based models such as DINOv2, offer powerful, generalized visual representations learned from extensive pretraining on large-scale datasets (13). These models have significantly enhanced performance across a wide array of vision tasks by providing prior knowledge that captures intricate image semantics without task-specific retraining (3; 11; 14). Yet, their capability in leveraging multi-modal information, particularly depth, for indoor semantic segmentation remains under-explored (15). Recent studies primarily focus on single-modal or RGB-centric scenarios, and only a few have begun integrating depth into VFM frameworks (3; 8; 16; 15). This limited exploration constitutes a gap in the current literature, leading to the idea of systematic investigation into multi-modal fusion strategies involving depth within VFM contexts.

In this paper, we revisit RGB-D fusion strategies by integrating the representational strengths of vision foundation models (15). Specifically, we explore and compare early, mid, and late fusion strategies to effectively incorporate depth information within the DINOv2 architecture (6; 10). We benchmark our approaches against the state-of-the-art U-Net-based model proposed by Tran et al. (8), critically assessing the advantages and limitations inherent to adopting VFMs for multimodal semantic segmentation. Following the comparative analysis, we further refine the best-performing model through targeted enhancement methods, aiming to establish a new standard in the effective utilization of depth data alongside foundational visual representations for indoor scene parsing (16).

II. RELATED WORK

In this section, we review prior work on RGB-D semantic segmentation along two dimensions: fusion strategies in convolutional models and adaptation regimes for vision foundation models (VFMs). We then identify the gap addressed by our three-stage experimental plan.

A. Fusion in Convolutional Encoder–Decoders

Early RGB-D segmentation methods extended standard CNNs by injecting depth via dual-stream architectures (7). Hazirbas *et al.* (5) process RGB and depth through parallel encoders and fuse their features after each block, outperforming naïve channel stacking (10). Building on the U-Net architecture (1), AdapNet++ uses self-supervised gating to adaptively re-weight RGB and depth, achieving a +6 pp mIoU gain on SUN RGB-D (4). More recently, DEADFL-UNet integrates EfficientNet and MobileNet encoders with a Super-Attention decoder. Fine-tuning the depth branch yields an additional +2 pp mIoU on NYUv2 compared to a frozen depth module (8). These works demonstrate that both the stage of fusion and selective parameter tuning can substantially impact performance (6; 15).

B. Side Adapters for Vision Transformers

Vision Transformers (ViTs) pretrained as VFMs offer rich priors for dense prediction. Guo *et al.* introduce the Heterogeneous Feature Integration Transformer (HFIT), a side-adapter for frozen DINOv2 or Depth-Anything backbones (3). HFIT comprises three modules: (i) a Duplex Spatial-Prior Extractor that produces multi-scale priors from RGB and relative depth predicted by the VFM; (ii) Re-calibrated Heterogeneous Feature Fusion blocks that inject these priors via confidence-weighted cross-attention; and (iii) a Holistic Gated Feature Integration layer before decoding. Adding only ~ 9 M trainable parameters, HFIT improves performance on Cityscapes mIoU from 80.2 % to 84.5 % and KITTI from 78.6 % to 79.9 %. Notably, HFIT is validated exclusively on outdoor driving benchmarks, so its applicability to cluttered indoor environments remains untested (3). Complementary work adapts DINOv2 with lightweight Low Rank Adaptation (LoRA) modules for robust BEV perception (16).

C. Adaptation Regimes for Vision Foundation Models

Beyond side adapters, several paradigms exist for adapting VFMs to segmentation tasks. A simple linear head on frozen DINOv2 rivals supervised baselines on NYUv2 (13), while the Segment Anything Model (SAM) demonstrates promptable segmentation at scale (17). Full fine-tuning of DINOv2, as shown by Keressies *et al.*, yields state-of-the-art robustness benchmarks with minimal learning rates (12; 11). Conversely, the LoRA technique adds only 1–3 M parameters to a frozen backbone and matches larger models by a $\times 30$ reduction in trainable weights (16). The Encoder-only Mask Transformer (EoMT) uses solely learnable queries on a frozen VFM to achieve 84 % mIoU on Cityscapes at real-time speeds (18). However, these studies typically omit or under-utilize aligned depth inputs.

D. Gap Analysis and Our Three-Stage Plan

While CNN-based fusion strategies and VFM adaptation regimes have both advanced RGB-D segmentation, systematic studies of how depth fusion interacts with VFM tuning in indoor scenes remain absent. To fill this gap, our work

proceeds in three stages. In Stage 0, we benchmark frozen versus fine-tuned DINOv2 backbones, each combined with an identical lightweight linear decoder, on NYUv2 dataset. In Stage 1, we fix the fine-tuned backbone and investigate early (input concatenation), mid (feature-level concatenation and projection), and late (LoRA and cross-modal attention) fusion strategies. Finally, Stage 2 refines the top-performing cross-modal attention model by pretraining the depth stream to emphasize geometric features, adopting a deeper CNN depth encoder, replacing naive upsampling with transposed convolutions in the decoder, and applying layer normalization before attention. All experiments adhere to the same data splits, augmentations, and compute budgets, with DEADFL-UNet serving as our primary CNN benchmark. This controlled evaluation yields comprehensive insight into where, and under which tuning regimes, depth enhances VFM-based indoor segmentation.

III. METHODOLOGY

A. RGB-Only Baselines

We implement two RGB-only segmentation baselines built on the DINOv2-ViT-L/14 backbone, both of which can be represented by Figure 1.

1) *Frozen Backbone with Linear Decoder:* Here, the ViT backbone is frozen end-to-end, yielding a zero-shot feature extractor. An input RGB image $\mathbf{x} \in \mathbb{R}^{3 \times H \times W}$ is first resized so H, W are multiples of the patch size $P = 14$. We feed it through DINOv2 to obtain patch-token embeddings

$$\mathbf{F} \in \mathbb{R}^{N \times D}, \quad D = 1024, N = (H/P)(W/P),$$

which we reshape to $(D, H/P, W/P)$. A single linear layer

$$R^D \xrightarrow{\mathbf{W} \in \mathbb{R}^{D \times 13}} R^{13}$$

applies per-patch classification, producing logits in $\mathbb{R}^{13 \times H/P \times W/P}$. Finally, we bilinearly upsample to (H, W) .

Trainable parameters: only the decoder weights $\mathbf{W}$ ($1024 \times 13 \approx 13,312$).

2) *Fully Fine-Tuned Backbone and Decoder:* To measure the upper bound of adaptation, we unfreeze the entire ViT backbone in addition to the decoder head. The same pre-processing and decoder architecture is used, but all ViT parameters are trained on the segmentation loss.

Trainable parameters: the decoder (≈ 13 K) plus all backbone weights (≈ 307 M), for a total of ≈ 307 M trainable parameters.

Both baselines share patch-aligned resizing and upsampling routines; the only difference is whether the backbone parameters remain frozen or are fully fine-tuned. Training hyperparameters, datasets, and evaluation metrics are detailed in §IV.

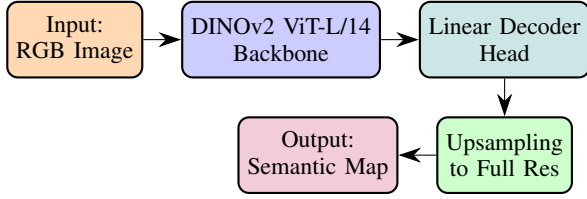


Fig. 1: Base RGB-only segmentation model: DINOv2 backbone with a linear decoder and upsampling to full resolution.

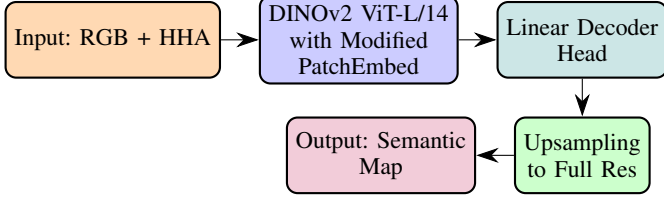


Fig. 2: Early fusion model: RGB and HHA depth concatenated at input, processed by DINOv2 with a modified patch embedding layer, followed by a linear decoder and upsampling.

B. Early Fusion via Input Concatenation

In the early fusion variant represented in Figure 2, we merge RGB and depth at the input level before any tokenization or Transformer blocks:

- **6-Channel Input:** We stack the three RGB channels and three HHA-encoded depth channels (Horizontal disparity, Height above ground, and the Angle with gravity) to form a tensor $\mathbf{X} \in R^{6 \times H \times W}$. This encoding captures richer geometric cues than raw depth alone and is also used in our CNN benchmark (8).
- **Modified Patch Embed:** We replace the original DINOv2 patch-embedding convolution

$$\text{Conv2d}(3 \rightarrow D, K = P, S = P)$$

with a new $\text{Conv2d}(6 \rightarrow D, K = P, S = P)$. Its weights are initialized by copying the pretrained RGB kernels into the first three input channels and setting the three HHA kernels to the mean of the RGB weights while biases are preserved.

- **Backbone & Decoder:** The fused tokens $\in R^{N \times D}$ pass through the standard DINOv2 ViT-L/14 blocks. A linear decoder head $R^D \rightarrow R^{13}$ is applied per token, and we bilinearly upsample the resulting logits to the original resolution (H, W) .
- **Training Regime:** We fully fine-tune the entire model, including the modified patch-embed convolution, all ViT blocks, and the decoder head.

Trainable parameters:

- Modified patch-embed conv: $\approx 6 \times 1024 \times 14^2 \approx 1.2\text{M}$
- Decoder head: $1024 \times 13 \approx 13\text{K}$
- Backbone: $\approx 307\text{M}$

Total: $\approx 308.2\text{M}$ parameters are trained.

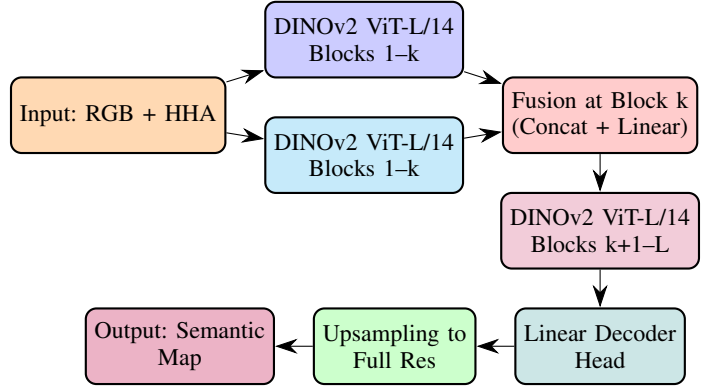


Fig. 3: Mid-fusion model: Separate RGB and depth streams up to fusion block k , fused via concatenation and linear projection, followed by joint transformer blocks, a linear decoder head, and upsampling.

C. Mid-Level Fusion via Feature Concatenation

In the mid-fusion variant, RGB and depth streams are processed independently up to an intermediate ViT block and then merged, as shown in Figure 3

- **6-Channel Input & Stream Splitting:** We take $\mathbf{X} \in R^{6 \times H \times W}$ (RGB + HHA) and split it into $\mathbf{X}_{\text{rgb}} \in R^{3 \times H \times W}$ and $\mathbf{X}_{\text{dep}} \in R^{3 \times H \times W}$.
- **Separate Embedding & Positional Encoding:** Each stream is passed through its own patch-embed + positional-encode yielding $\mathbf{T}_{\text{rgb}}, \mathbf{T}_{\text{dep}} \in R^{N \times D}$, where $N = (H/P)(W/P)$, $D = 1024$.
- **Independent Blocks (1-k):** We run both $\mathbf{T}_{\text{rgb}}$ and $\mathbf{T}_{\text{dep}}$ through the first $k = L/2$ transformer blocks (of the $L = 24$ total).
- **Fusion at Block k :** We concatenate along the channel axis,

$$[\mathbf{T}_{\text{rgb}}; \mathbf{T}_{\text{dep}}] \in R^{N \times 2D},$$

then project back to $R^{N \times D}$ via a trainable $R^{2D} \rightarrow R^D$ layer.

- **Joint Blocks ($k + 1 - L$):** The fused tokens traverse the remaining $L - k$ ViT blocks and final layer-norm.
- **Decoder & Upsampling:** We reshape the output $\in R^{N \times D}$ to a $(D, H/P, W/P)$ feature map, apply a per-token linear head $R^D \rightarrow R^{13}$, and bilinearly upsample to (H, W) .

Trainable parameters:

- RGB encoder (all $L = 24$ blocks, patch-embed, pos-embed): $\approx 307\text{M}$
- Depth encoder (only the first $k = 12$ blocks, plus its patch-embed & pos-embed): $\approx 153.5\text{M}$
- Fusion projection ($2D \rightarrow D$): $2 \times 1024 \times 1024 \approx 2.1\text{M}$
- Decoder head ($D \rightarrow 13$): $1024 \times 13 \approx 13\text{K}$

Total: $\approx 462.6\text{M}$ parameters are fine-tuned.

D. Late Fusion via Adapter Modules

In the late-fusion setting, RGB and depth streams are processed independently by separate encoders, and their features are merged only via a lightweight adapter module immediately before the segmentation head. In both variants below, the DINOv2 backbone is *fully fine-tuned* along with the adapter and decoder. Figure 4 illustrates the architecture.

1) *Depth-Conditioned LoRA Adapter*: We split the 6-channel input $\mathbf{X} \in \mathbb{R}^{6 \times H \times W}$ into RGB $\mathbf{X}_{\text{rgb}} \in \mathbb{R}^{3 \times H \times W}$ and HHA-encoded depth $\mathbf{X}_{\text{dep}} \in \mathbb{R}^{3 \times H \times W}$.

- **RGB Stream**: $\mathbf{X}_{\text{rgb}}$ passes through the DINOv2 ViT-L/14 encoder (all $L = 24$ blocks fine-tuned), yielding $\mathbf{T}_{\text{rgb}} \in \mathbb{R}^{N \times D}$, $D = 1024$.
- **Depth Stream**: $\mathbf{X}_{\text{dep}}$ is fed into a small CNN-based HHA encoder (two 3×3 conv layers, global pooling, linear) producing $\mathbf{d} \in \mathbb{R}^D$.
- **LoRA Fusion**: We apply a depth-conditioned LoRA module:

$$\Delta(\mathbf{T}_{\text{rgb}}, \mathbf{d}) = \frac{\alpha}{r} \mathbf{B}(\mathbf{A}(\mathbf{T}_{\text{rgb}} + \mathbf{d})), \quad r = 4, \alpha = 1,$$

and set $\mathbf{T}_{\text{fused}} = \mathbf{T}_{\text{rgb}} + \Delta$.

- **Decoder & Upsampling**: A per-token linear head $\mathbb{R}^D \rightarrow \mathbb{R}^{13}$ produces logits, which are bilinearly upsampled to (H, W) .

Trainable parameters:

- DINOv2 backbone (ViT-L/14): $\approx 307\text{M}$
- HHA encoder (CNN+linear): $\approx 0.09\text{M}$
- LoRA adapter ($\mathbf{A}, \mathbf{B}$, depth projection): $\approx 1.06\text{M}$
- Decoder head ($D \rightarrow 13$): $\approx 0.01\text{M}$

Total: $\approx 308.2\text{M}$ trainable parameters.

2) *Cross-Modal Attention (CMA) Adapter*: Again, split $\mathbf{X}$ into $\mathbf{X}_{\text{rgb}}$ and $\mathbf{X}_{\text{dep}}$, but now encode depth via a patch-based Transformer:

- **RGB Encoder**: $\mathbf{X}_{\text{rgb}} \rightarrow \text{DINOv2 ViT-L/14 (fully fine-tuned)} \rightarrow \mathbf{T}_{\text{rgb}} \in \mathbb{R}^{N \times D}$.
- **Depth Patch Encoder**: $\mathbf{X}_{\text{dep}} \rightarrow \text{three-stage Conv2d stack} \rightarrow \mathbf{T}_{\text{dep}} \in \mathbb{R}^{N \times D}$.
- **Cross-Modal Attention**:

$$\mathbf{T}_{\text{fused}} = \text{Attn}(\mathbf{T}_{\text{rgb}}, \mathbf{T}_{\text{dep}}, \mathbf{T}_{\text{dep}}) + \mathbf{T}_{\text{rgb}},$$

with 8 heads.

- **Decoder & Upsampling**: As above, a linear head $\mathbb{R}^D \rightarrow \mathbb{R}^{13}$ and bilinear upsampling yield the final semantic map.

Trainable parameters:

- DINOv2 backbone (ViT-L/14): $\approx 307\text{M}$
- Depth patch encoder (Conv2d stack): $\approx 28\text{M}$
- Cross-attention adapter (Q/K/V + out projection): $\approx 4.2\text{M}$
- Decoder head ($D \rightarrow 13$): $\approx 0.01\text{M}$

Total: $\approx 339.2\text{M}$ trainable parameters.

All late-fusion variants follow the pattern: independent full fine-tuning of encoders $\rightarrow$ lightweight fusion adapter $\rightarrow$ linear decoding.

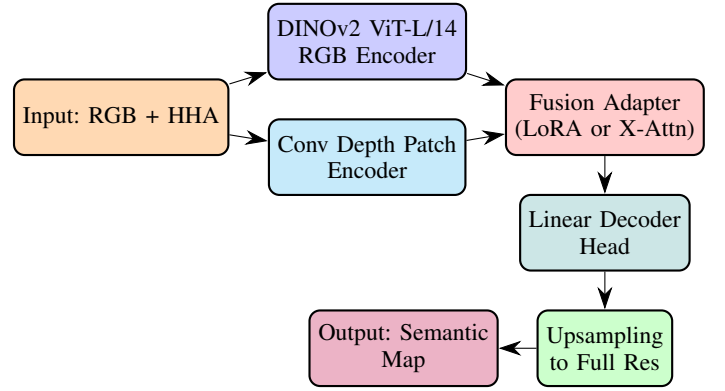


Fig. 4: Late fusion models (LoRA and Cross-Modal Attention): separate RGB and depth streams, fusion adapter combining features, followed by a linear decoder head and upsampling.

E. Architectural Refinements for Cross-Modal Attention Fusion

We introduced four progressive enhancements to the cross-modal attention fusion model to improve depth utilization, training stability, and spatial fidelity. First, we employ a progressive freezing schedule: during the first 10 epochs all parameters of the DINOv2 backbone and the decoder are frozen, allowing only the depth encoder and the fusion adapter to update. This encourages the network to focus on learning geometric features from the HHA-encoded depth channels. After epoch 10, we unfreeze every module and continue full fine-tuning for the remainder of training.

Second, we replace the original shallow HHA encoder with a deeper convolutional network that projects the three HHA channels into the ViT embedding dimension. Concretely, the new depth encoder consists of a 7×7 convolution (stride 2, padding 3) into D channels, a 3×3 convolution (stride 1, padding 1), and a final convolution with kernel and stride equal to the patch size P . Each convolution is followed by BatchNorm and GELU. This deeper stack extracts richer geometric cues prior to fusion.

Third, since multi-head attention is sensitive to feature scale and distribution, we insert a learnable LayerNorm on both the RGB token sequence and the depth token sequence immediately before cross-attention. Formally, letting T_{rgb} and T_{dep} denote the pre-fusion tokens, we compute

$$\tilde{T}_{\text{rgb}} = \text{LN}(T_{\text{rgb}}), \quad \tilde{T}_{\text{dep}} = \text{LN}(T_{\text{dep}}),$$

and feed $\tilde{T}_{\text{rgb}}$ and $\tilde{T}_{\text{dep}}$ into the cross-attention adapter. This normalization stabilizes attention weights and yields more consistent fused representations.

Finally, to improve upsampling fidelity and boundary delineation, we replace the per-token linear decoder with a multi-stage transposed-convolution decoder. This decoder first projects the feature map from D to 512 channels via a 3×3 Conv2d+ReLU, then applies three ConvTranspose2d layers

(each doubling spatial resolution and followed by Batch-Norm+ReLU), and concludes with a 1×1 Conv2d to produce class logits. Any minor resolution mismatch is corrected by a final bilinear interpolation.

These refinements add a moderate number of parameters—tens of millions from the deeper depth encoder, two small LayerNorms ($2 \times D$ learnable scales and biases), and roughly 5M parameters in the ConvTranspose decoder—resulting in under a 10% increase in the total trainable parameter count relative to the base cross-modal attention fusion model. All other training protocols and hyperparameters remain as detailed in §IV.

IV. EXPERIMENTAL SETUP

A. Dataset

We conduct all experiments on the NYUv2 dataset (9). Following Tran *et al.* (8), we keep the official 795/654 train/test split. The evaluation is carried out on the *13-class protocol*: *Bed, Books, Ceiling, Chair, Floor, Furniture, Object, Painting, Sofa, Table, TV, Wall, Window* plus a single *Unlabeled* category that is ignored when computing metrics.

a) Depth representation.: Raw metric depth frames are converted off-line to the HHA (horizontal disparity, height above ground, angle with gravity) encoding using the public implementation of Chen¹ and cached as 3-channel PNGs. Figure 5 contains examples of the data in the original dataset and HHA encoded depth. These HHA maps are stacked with the RGB images according to the fusion strategies under test (Section III).

b) Input resolution and augmentations.: All models receive images at the native NYUv2 resolution (480×640). We apply the same data-augmentation pipeline across models:

- **Random horizontal flip** ($p = 0.5$);
- **Color-jitter** on brightness, contrast, saturation and hue;
- **Random scaling** in the range $[0.5, 1.2]$ followed by zero-padding and a random crop back to 480×640 .

These result in samples such as the ones in Figure 6

c) Class imbalance.: The dataset exhibits a pronounced long-tail distribution (Table I). To compensate, we employ the class-balanced focal loss described in Section IV-B0c.

B. Experimental Framework

a) Scope.: Architectural details of the RGB baselines, fusion variants and the *best-model improvement* study are provided in Section III. The present subsection specifies the common training protocol that makes all comparisons fair and reproducible.

b) Optimisation and schedule.: All models are trained for 40 epochs with a *poly* learning-rate schedule $LR(t) = LR_0(1 - \frac{t}{T})^{0.9}$, where T is the total number of iterations. We employ the AdamW optimiser ($\beta_1=0.9$, $\beta_2=0.999$, weight-decay 0.05) and a batch size of 16. Different modules start from distinct base learning rates:

¹<https://github.com/charlesCXX/Depth2HHA-python>

TABLE I: Class distribution in the NYUv2 13-class split.

Class	Frequency (%)
Unlabeled	17.61
Bed	3.51
Books	0.54
Ceiling	1.45
Chair	3.38
Floor	9.81
Furniture	14.60
Object	12.88
Painting	1.99
Sofa	2.35
Table	3.01
TV	0.55
Wall	23.57
Window	4.75

Module	RGB encoder	Depth encoder	Decoder / Adapter
LR ₀	1×10^{-5}	1×10^{-4}	1×10^{-4}

c) Loss function.: Training minimises a **class-balanced focal loss** that merges the focusing term of Focal Loss (19) (with $\gamma = 2$) and inverse-frequency weights to counteract the severe class imbalance of NYUv2 (Table I). For a pixel with ground-truth label y and predicted probability p_y , the loss is

$$\mathcal{L} = -w_y (1 - p_y)^\gamma \log p_y, \quad w_y = \frac{\frac{1}{f_y + \varepsilon}}{\sum_{c=0}^{C-1} \frac{1}{f_c + \varepsilon}} C,$$

where f_y is the empirical pixel frequency of class y , $C = 13$ and $\varepsilon = 10^{-6}$ for numerical stability.

An implementation is released with the code repository for reproducibility.

d) Fairness and reproducibility.: The optimiser, schedule, batch size and loss are *identical* for every model so that observed performance differences stem solely from architectural or fusion choices.

C. Implementation Details

a) Hardware and software stack.: All experiments were executed on a single NVIDIA A100 80 GB GPU running Ubuntu 22.04. The codebase uses Python 3.9.16 and PyTorch 2.5.1 without additional high-level training frameworks.

b) Mini-batch construction.: An *effective* batch size of **16** images is obtained by processing micro-batches of **8** images on-device and accumulating gradients for **2** optimisation steps before each parameter update. Automatic mixed precision and gradient checkpointing are disabled to keep numerical behaviour identical across models.

c) Data loading.: RGB frames and pre-computed HHA depth encodings are stored as PNGs on Network Attached Storage (NAS). Data are streamed with eight `torch.utils.data.DataLoader` workers (`pin_memory=True`, `prefetch_factor=2`).

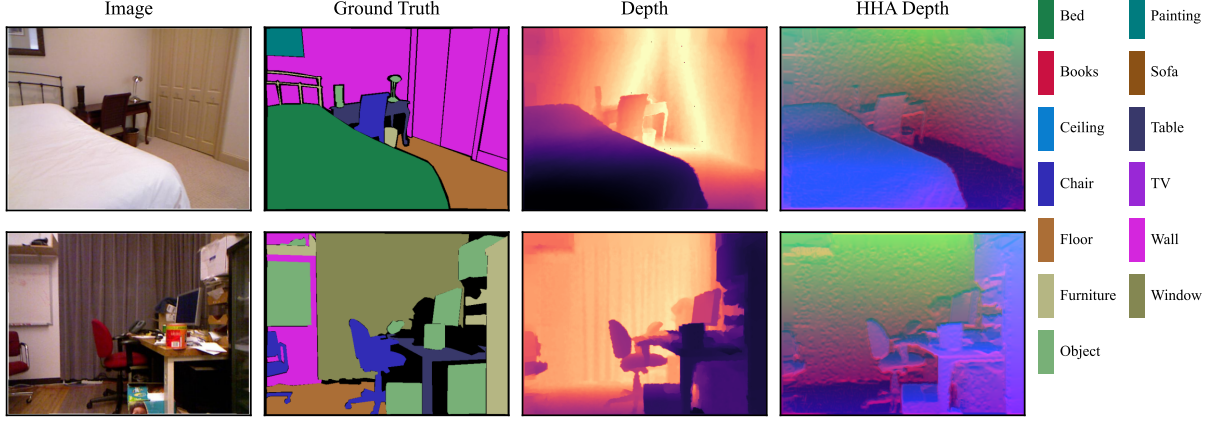


Fig. 5: NYUv2 dataset samples and class labels

d) *Checkpointing and experiment tracking*: Metrics (training/validation loss, mIoU, mean accuracy, pixel accuracy, frequency-weighted IoU, AUROC and expected calibration error) are logged with Weights&Biases. The checkpoint corresponding to the *lowest validation loss* is saved and later used for test-set evaluation.

e) *Reproducibility*: A global pseudo-random seed of 42 is set for Python, NumPy and PyTorch. All remaining hyper-parameters (optimiser, learning- rate schedule, loss, etc.) follow the common protocol specified in Section IV-B.

D. Evaluation Metrics

Model performance is assessed with a mix of recognition, efficiency, and calibration criteria computed on the standard NYUv2 test set. All metrics are derived from the confusion matrix $\mathbf{C} \in \mathbb{N}^{(C-1) \times (C-1)}$ obtained after removing the *Unlabeled* class (`ignore_index=0`).

- **Pixel Accuracy (PA)**. Ratio of correctly classified pixels to the total number of pixels:

$$\text{PA} = \frac{\sum_i C_{ii}}{\sum_{i,j} C_{ij}}$$

- **Mean Accuracy (mAcc)**. Average of per-class accuracies

$$\text{mAcc} = \frac{1}{C-1} \sum_i \frac{C_{ii}}{\sum_j C_{ij}}$$

- **Mean Intersection-over-Union (mIoU)**. The primary benchmark metric, computed as the macro average of class IoUs:

$$\text{mIoU} = \frac{1}{C-1} \sum_i \frac{C_{ii}}{C_{ii} + \sum_j C_{ij} - C_{ii} + \sum_j C_{ji} - C_{ii}}$$

- **Frequency-Weighted IoU (FWIoU)**. Re-weights each class IoU by its pixel frequency to reflect the global error rate.

- **Expected Calibration Error (ECE)**. Measures the gap between predicted confidence and empirical accuracy over $n=10$ equal-width bins of pixel confidences:

$$\text{ECE} = \sum_{b=1}^n |\text{acc}(b) - \text{conf}(b)| \frac{|b|}{N}$$

where $|b|$ is the number of pixels in bin b and N the total number of valid pixels.

- **Pixel-level AUROC**. Area under the ROC curve obtained by treating the predicted confidence as a binary classifier of correctness (20). AUC closer to 1 indicates better separation between correct and incorrect pixels.

V. RESULTS

A. Training and Validation Dynamics

Figures 7–9 depict the evolution of loss, mIoU and pixel accuracy over 40 training epochs. Figure 7 contrasts the *Frozen* and *Fine-tuned* RGB-only backbones, while Figure 8 compares the four fusion strategies. The *Mid-Fusion* variant exhibits slower convergence and lower plateau values across all three metrics, whereas both late-fusion adapters track the fine-tuned RGB baseline closely. Figure 9 evidences steady gains from each architectural refinement applied to the cross-modal attention model, without signs of severe over-fitting.

B. Quantitative Performance on NYUv2

Table II reports the pixel-wise performance of all evaluated architectures on the NYUv2 13-class benchmark. Metrics include mean Intersection-over-Union (mIoU), frequency-weighted IoU (FWIoU), pixel accuracy (PA), mean class accuracy (mAcc), expected calibration error (ECE) and pixel-level AUROC. The *Cross-Modal Attention Fusion* model attains the highest mIoU of 71.37 %, surpassing the depth-augmented CNN baseline (DEADFL-UNet) by more than 12.5 percentage

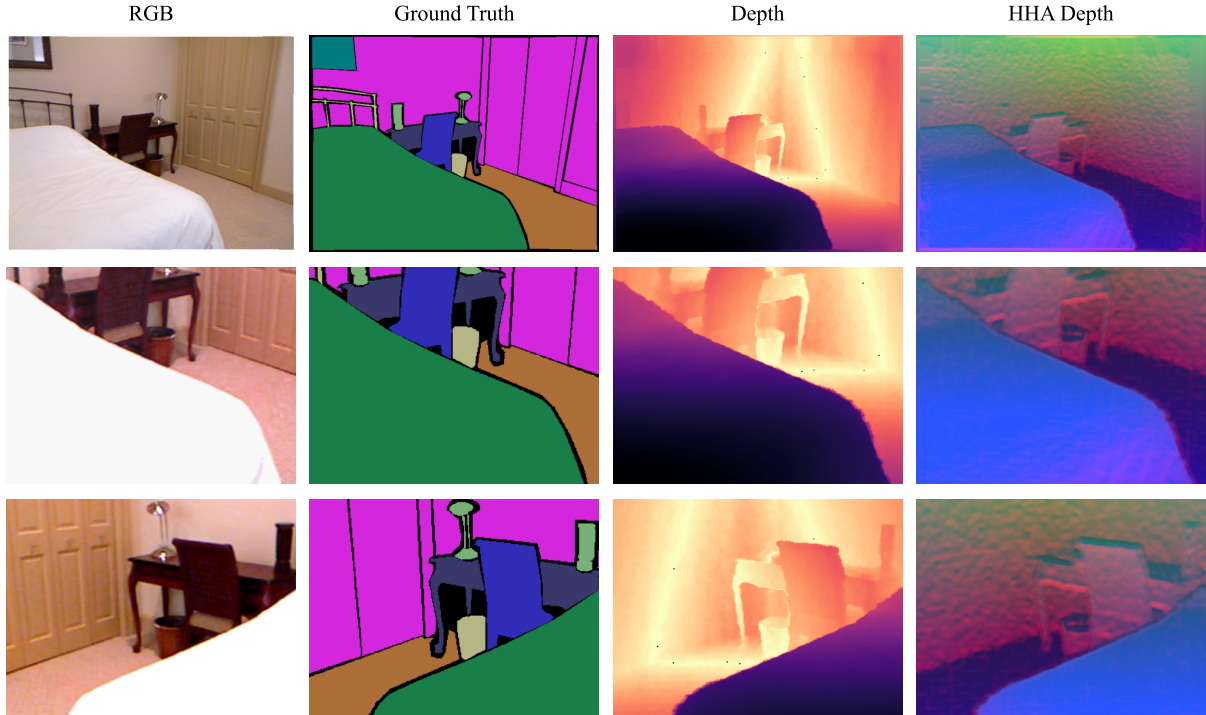


Fig. 6: Data augmentation samples obtained from the original data in the first row.

points. Successive ablations demonstrate that freezing schedules, deeper depth encoders and a transpose-convolutional decoder progressively lift performance, culminating in a best mIoU of 76.31%.

C. Qualitative Results

Representative segmentation maps are shown in Figure 10. Elements correspond to the RGB input, HHA depth, ground-truth annotations, the fine-tuned RGB-only baseline, the *Cross-Modal Attention* model, and the final refined model, respectively. Visual inspection confirms progressive reductions in wall-floor bleeding and finer delineation of thin structures (e. g. chair legs and tabletop edges).

VI. DISCUSSION

A. A Strong DINOv2 Baseline

Using a *fully frozen* DINOv2 ViT-L/14 single trained linear decoder already matches the mIoU reported for the depth-augmented U-Net of Tran *et al.* on NYUv2 (Table II). Fine-tuning the backbone lifts performance by ≈ 10 percentage points, establishing a markedly stronger RGB reference than earlier CNN-based pipelines.² This result underscores how much representational capacity VFMs contribute, even *without* depth input.

²We treat “frozen” and “fine-tuned” backbones as separate configurations to isolate the value added by large-scale pre-training.

B. Why Naïve Fusion Falls Short

Early fusion through simple channel concatenation at the input, offers no measurable benefit over the RGB-only baseline, while mid fusion degrades mIoU (Fig. 8). The drop can be traced to our depth pathway: it re-uses the first DINOv2 ViT block trained on RGB images, and although that block is *not* frozen, the allowed optimisation is insufficient to learn depth-specific primitives from scratch. Only the cross-modal attention (CMA) head, which conditions RGB features *with* depth rather than merely blending them, yields a tangible gain, confirming that feature-level interaction is necessary once strong semantic priors are already present in the backbone.

C. Targeted Unfreezing and Architectural Tweaks

Letting the depth encoder learn in isolation for the first ten epochs reduces modality imbalance and adds a further 2.3 pp mIoU—an effect also observed in depth-adaptive schedules for CNN backbones (8). Deeper convolutional depth encoders contribute finer geometric detail, and replacing bilinear up-sampling with a transpose-convolutional decoder removes checkerboard artefacts, together accounting for another percentage point. Overall, these incremental refinements compound to a meaningful boost without inflating parameter count by more than 5%.

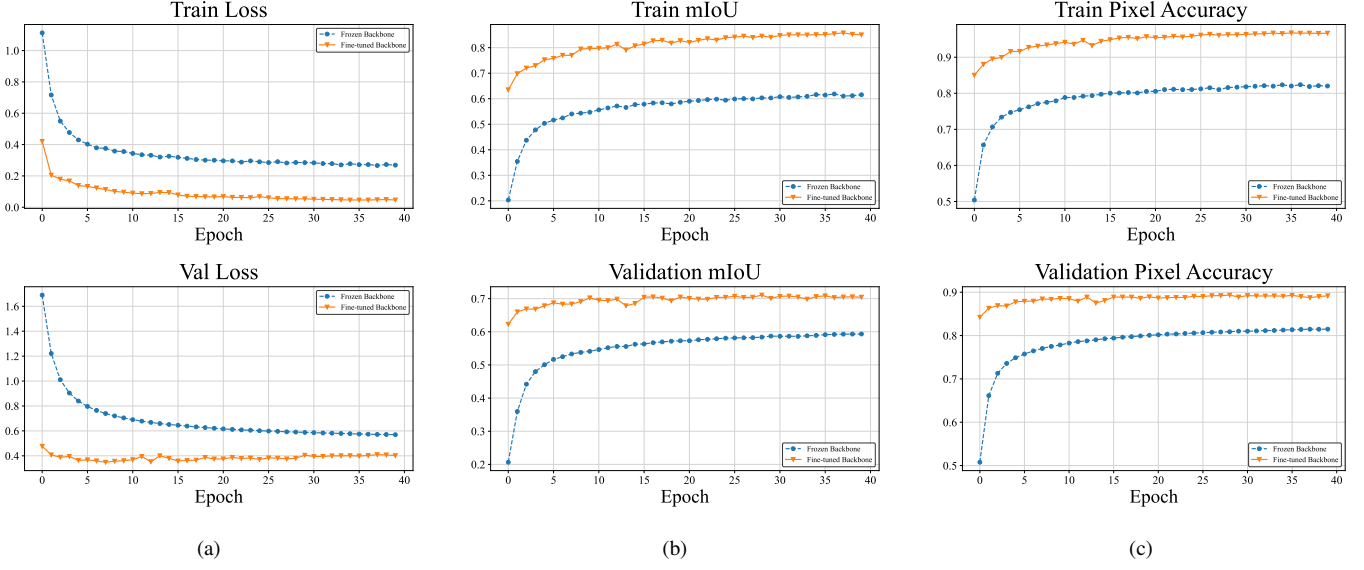


Fig. 7: Train (top row) and validation (bottom row) curves over epochs for three metrics: (a) Loss, (b) mIoU, (c) Pixel Accuracy for the RGB baseline models.

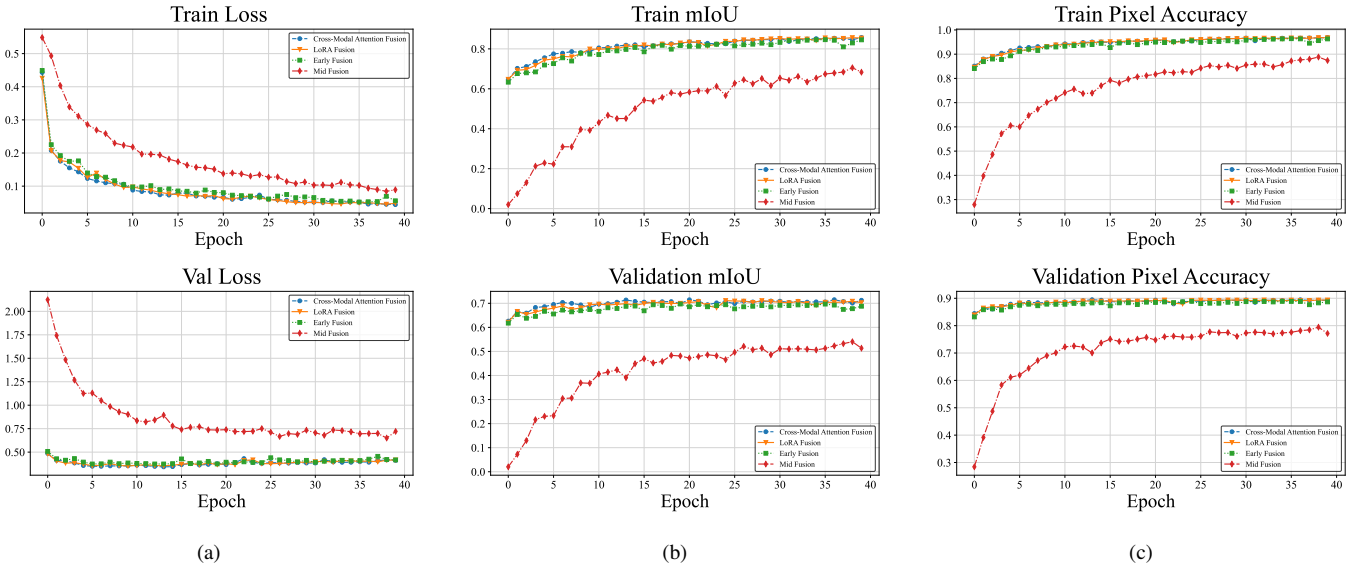


Fig. 8: Train (top row) and validation (bottom row) curves over epochs for three metrics: (a) Loss, (b) mIoU, (c) Pixel Accuracy for the diffent fusion models

D. Training Dynamics and Generalisation

Loss and accuracy curves (Figures 7–9) show that CMA converges at roughly the same pace as the RGB fine-tuned model despite its additional branch. Moreover, the validation-train gap narrows once the depth-only unfreeze schedule is applied, suggesting that uncontrolled early sharing of depth and RGB gradients can over-fit ViT layers to noisy depth pixels. These observations align with consistency-regularised RGB-D frameworks reported for LuSeg (21).

E. Qualitative Insights

Qualitative maps (Fig. 10) confirm that depth guidance is most impactful on wall-floor junctions and thin structures such as chair legs—regions where RGB cues are either ambiguous or texturally weak.

VII. CONCLUSION AND FUTURE WORK

This work revisited indoor RGB-D semantic segmentation under the lens of Vision Foundation Models. We showed that a frozen DINOv2 backbone with a lightweight linear decoder matches the depth-aware U-Net of Tran *et al.*, while fine-tuning the backbone yields an additional ~ 10 pp jump in

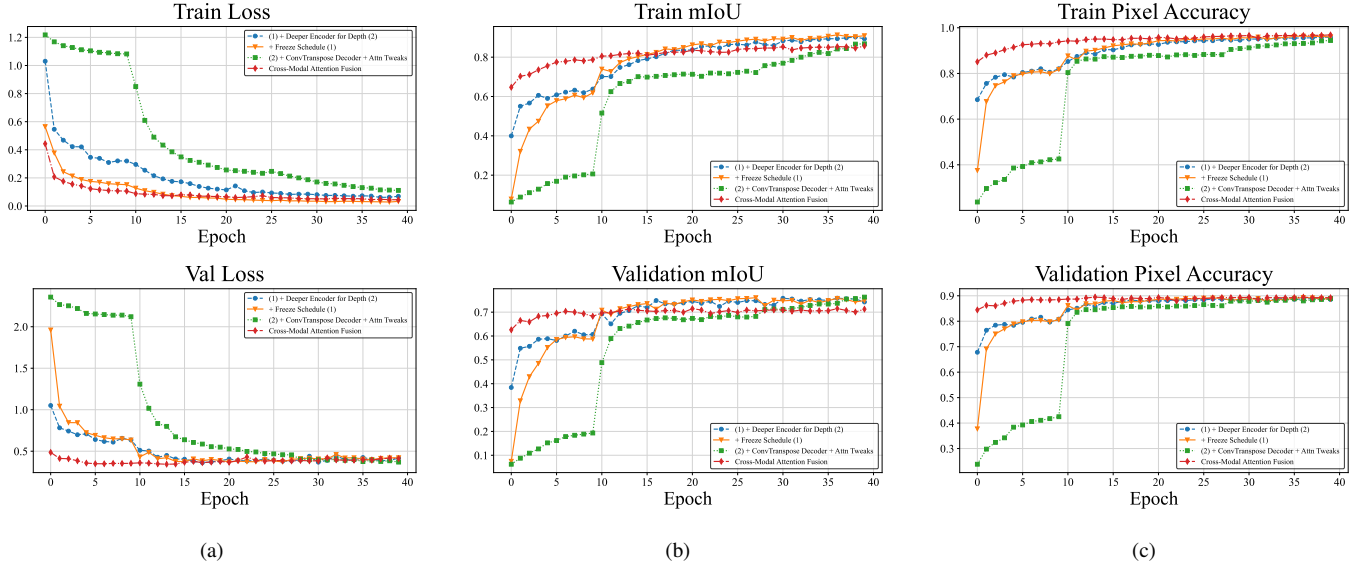


Fig. 9: Train (top row) and validation (bottom row) curves over epochs for three metrics: (a) Loss, (b) mIoU, (c) Pixel Accuracy for the Cross Modal Attention Model at different stages of architectural tweaks.

TABLE II: Semantic segmentation performance across models

Model	mIoU	FWIoU	PA	mAcc	ECE (%)	Pixel-Level AUROC
DEADFL-UNet (8)	58.73	60.78	67.02	72.51	-	-
Frozen Backbone	59.34	69.20	81.47	69.25	2.32	0.8205
Fine-tuned Backbone	69.85	80.43	88.90	77.45	3.67	0.8717
Early Fusion	69.71	80.05	88.57	78.66	4.07	0.8618
Mid Fusion	54.04	67.02	79.44	65.59	6.01	0.8316
LoRA Fusion	69.85	80.55	88.95	78.15	3.94	0.8716
Cross-Modal Attention Fusion	71.37	81.43	89.48	79.90	3.82	0.8896
+ Freeze schedule	75.75	80.88	89.12	85.25	4.23	0.8632
+ Deeper Encoder for Depth	75.86	80.34	88.77	86.48	5.42	0.8635
+ ConvTranspose Decoder w/ Attn Tweaks	76.31	80.59	88.94	86.76	1.80	0.8558

mIoU. Simple depth concatenation (early fusion) and mid-fusion copies of the RGB encoder fail to exploit geometric cues, but a *cross-modal attention* adapter recovers a clear advantage. Further targeted refinements add a residual performance boost. Together, these findings underline two messages: (i) large-scale pre-training provides an exceptionally strong RGB baseline; and (ii) depth remains useful, but only when fused through mechanisms capable of learning feature-level conditioning.

Promising next steps include

- **Depth-to-RGB distillation:** transferring geometric priors into the RGB stream at training time to remove the runtime cost of a dedicated depth branch;
- **Self-supervised depth:** replacing metric sensors with relative depth from models such as *Depth-Anything* (22), broadening applicability to monocular mobile devices;
- **Robustness and scale:** evaluating on larger, noisier datasets (e.g. ARKitScenes (23)) and studying failure modes under missing-pixel or time-of-flight noise.

These directions aim to make RGB-D fusion both cheaper and, paving the way for deployment in resource-constrained

platforms.

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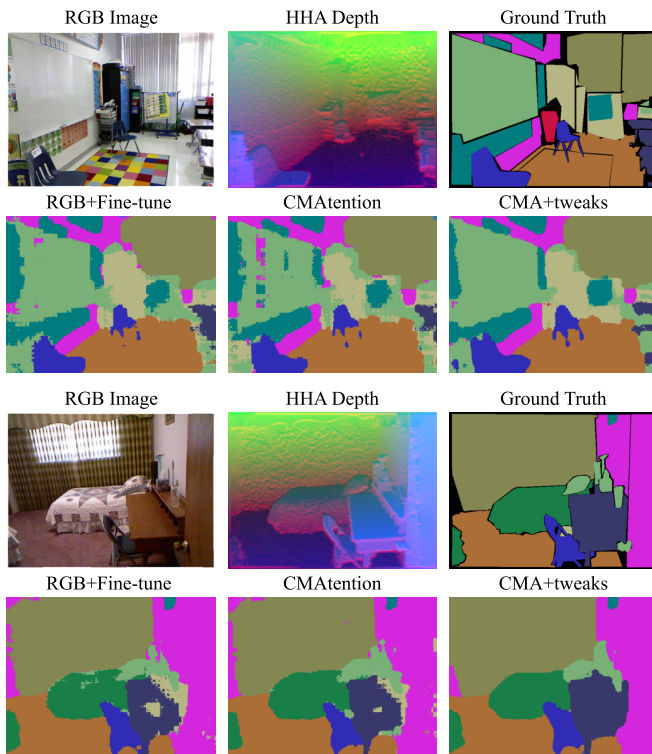


Fig. 10: Segmentation performance comparison between the Fine-tuned RGB baseline, Cross Modal Attention Fusion model with and without architectural changes to boost performance.

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